ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE C · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Electronics and Digital Concepts for Robotics paper 1 · practice c.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable diagrams.

SECTION A

QUESTION 1

1 1.1 D ✓ 1.2 A ✓ 1.3 C ✓ 1.4 B ✓ 1.5 A ✓ 1.6 D ✓ 1.7 B ✓ 1.8 C ✓ 1.9 A ✓ 1.10 B ✓ (10)

[10]

QUESTION 2

2 2.1 True ✓ 2.2 False ✓ 2.3 False ✓ 2.4 True ✓ 2.5 True ✓ 2.6 False ✓ 2.7 True ✓ 2.8 True ✓ 2.9 False ✓ 2.10 True ✓ (10)

[10]

QUESTION 3

3 3.1 C ✓ 3.2 I ✓ 3.3 F ✓ 3.4 B ✓ 3.5 H ✓ 3.6 D ✓ 3.7 E ✓ 3.8 J ✓ 3.9 A ✓ 3.10 G ✓ (10)

[10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1.1 $X_L = 2\pi fL = 2\pi \times 50 \times 0{,}2$ ✓ $= 62{,}83\ \Omega$ ✓ (2)

4.1.2 $Z = \sqrt{(R^2 + {X_L}^2)}$ ✓ $= \sqrt{(40^2 + 62{,}83^2)}$ ✓ $= 74{,}48\ \Omega$ ✓ (3)

4.1.3 $I = V \div Z = 230 \div 74{,}48$ ✓ $= 3{,}09$ A ✓ (2)

4.2 Any THREE ✓✓✓: light dimmers; AC motor speed control (fans, drills); AC switching / solid-state relays; heater and temperature control; soft starters. (3)

[10]

QUESTION 5

5.1 Any TWO losses with how they are reduced, ✓✓ each: copper (I^2R) losses in the windings – thicker conductors / low-resistance copper; eddy-current losses in the core – laminated core with insulated laminations; hysteresis losses – core of silicon steel (soft magnetic material). (4)

5.2 $E_T = 6 \times 1{,}5 = 9\ V$ ✓; $r_T = 6 \times 0{,}1 = 0{,}6\ \Omega$ ✓; $I = 9 \div (5{,}4 + 0{,}6)$ ✓ $= 1{,}5\ A$ ✓; $V = 1{,}5 \times 5{,}4$ ✓ $= 8{,}1\ V$ ✓ (6)

[10]

QUESTION 6

6.1 An amplifier ✓; positive feedback ✓; a frequency-determining network (RC or LC / crystal) ✓ (3)

6.2 $f = 1{,}44 \div ((R_1 + 2R_2) \times C)$ ✓ $= 1{,}44 \div ((1\,000 + 20\,000) \times 10 \times 10^{-6})$ ✓✓ $= 6{,}86\ Hz$ ✓ (4)

6.3 Any THREE ✓✓✓: timers / time delays; switch debouncing; pulse stretching; missing-pulse detectors; touch switches; one-shot triggers for alarms. (3)

[10]

QUESTION 7

7.1 $R = (V_{in} - V_Z) \div (I_Z + I_L)$ ✓ $= (12 - 5{,}1) \div 0{,}025$ ✓ $= 276\ \Omega$ ✓ (3)

7.2 When the supply rises, the extra voltage appears across the series resistor ✓; the Zener conducts more current ✓, so the current through the series resistor rises and the voltage drop across it increases, keeping the voltage across the Zener and load at about 5,1 V ✓. (3)

7.3 Set the meter to the diode test range ✓; red probe on the anode and black on the cathode – a good silicon diode reads about 0,6–0,7 V ✓; reverse the probes – a good diode shows OL (open) ✓; a reading near 0 in both directions means it is shorted, OL in both directions means it is open (faulty) ✓. (4)

[10]

QUESTION 8

8.1 $V_p = 3{,}5 \times 2$ ✓ $= 7\ V$ ✓ (2)

8.2 $V_{p\text{-}p} = 2 \times 7$ ✓ $= 14\ V$ ✓ (2)

8.3 $V_{rms} = 0{,}707 \times 7$ ✓ $= 4{,}95\ V$ ✓ (2)

8.4 $T = 4 \times 0{,}5$ ✓ $= 2\ ms$ ✓ (2)

8.5 $f = 1 \div 0{,}002$ ✓ $= 500\ Hz$ ✓ (2)

[10]

QUESTION 9

9.1 The complement of a sum equals the product of the complements: $\overline{A + B} = \overline{A} \cdot \overline{B}$ ✓✓. The complement of a product equals the sum of the complements: $\overline{A \cdot B} = \overline{A} + \overline{B}$ ✓✓. (4)

9.2 $F = A(B + \overline{B})$ ✓ $= A \cdot 1 = A$ ✓ (2)

9.3 Two-input OR gate with inputs A and B ✓; NOT gate on C ✓; two-input AND gate ✓ combining the OR output and $\overline{C}$, output labelled F ✓ (4)

[10]

QUESTION 10

10.1 A separately excited generator gets its field current from an external DC source ✓✓. A self-excited generator supplies its own field current from its armature (shunt, series or compound connection) ✓✓, starting from residual magnetism. (4)

10.2 $E_b = V - I_a R_a$ ✓ $= 220 - 20 \times 0{,}5$ ✓ $= 210\ V$ ✓ (3)

10.3 With no load, the armature current and field flux are very small ✓; the speed of a series motor rises as the flux falls ✓, so without a load it can race to a dangerously high speed and damage itself ✓. (3)

[10]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE